A Speech Encryption Algorithm Based on Blind Source Separation

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Abstract--**In this paper, we present a novel speech encryption algorithm based on blind source separation (BSS). Our approach integrates a modified time domain scrambling scheme with an amplitude scrambling method which masks the speech signal with a random noise by specific mixing. The resulting system can securely encrypt the speech files for the purpose of storing speech messages and transmitting them over the Internet. There are two major advantages associated with this system. The first advantage is that it makes the encrypted speech sound like white noise. The second advantage is that it does not impose any restriction on the key space. Our system is systematically evaluated, and it shows a high level of security with excellent audio quality.**

Keywords: Blind source separation, independent component analysis, speech encryption.

I. INTRODUCTION

AS speech communications become more and more widely used and even more vulnerable, the importance of providing a high level of security is dramatically increasing. As such, a variety of the speech encryption techniques has been introduced. The analogue encryption has been one of the popular encryption techniques widely used in speech communication. Along this line, there are three main categories: frequency domain scrambling such as the frequency inverter and the band splitter, time domain scrambling such as the time element scrambling, and two-dimensional scrambling combining the frequency domain scrambling with the time domain scrambling [1]. Besides, there is an amplitude scrambling technique, also known as the masking, which covers the speech signal by the linear addition of pseudo-random amplitudes [1]. Analogue speech encryption in the transform domain, e.g., fast Fourier transform, discrete cosine transform and wavelet transform, etc. [2]–[4] has also been developed. Recently, some new speech encryption methods have been proposed, e.g. chaotic cryptosystem [5], encryption using circulant transformations [6], etc. In this paper, we explore a blind source separation based speech encryption algorithm.

Blind source separation (BSS) is a technique of array signal processing that aims to recover a set of unknown mutually independent source signals from their observed mixtures without knowing the mixing coefficients. Independent component analysis (ICA) is a special case of BSS with linear and instantaneous mixture. BSS has promising applications to communications, biomedical engineering, etc. [7]–[10]. Its applications for signal encryption thus far have been mostly focused on the image encryption. For example, Kasprzak and Cichocki [11] proposed an application of ICA for hiding a single image transmitted by communication channels. We have recently applied BSS to add another encryption level besides the existing encryption methods for image cryptosystems in [12]. By using the separation-keys, we further proposed a mixed masking image encryption based on BSS [13]. Motivated by our recent work in the image encryption, we here present a BSS based speech encryption algorithm used for encrypting speech files. The algorithm effectively combines two encryption techniques, the time element scrambling method and the masking method, into one integrated system by virtue of the BSS. As a result, the system has substantially improved security performance while still retaining excellent audio quality.

This paper is organized as follows. In Section II, we give the mixing model of BSS. Sections III-IV, respectively, explain the BSS encryption and decryption of the system in detail. Section V provides evaluation experiments and shows the results. Finally, we conclude the paper in Section VI.

II. BSS MIXING MODEL

Suppose that there exist M independent source signals $s_1(t),\ldots,s_M(t)$ and N observed mixtures $x_1(t),\ldots,x_N(t)$ of the source signals (usually $N \geq M$). The simple BSS mixing model assumes that these mixtures are linear, instantaneous, and noiseless, which can be represented compactly by the following mixing equation:

$$\mathbf{x}(t) = \mathbf{A}\mathbf{s}(t) \tag{1}$$

where $\mathbf{s}(t) = [s_1(t),\ldots,s_M(t)]^T$, which is an $M \times 1$ column vector collecting the source signals, vector $\mathbf{x}(t)$ similarly collects the N observed (mixed) signals, and $N \times M$ mixing matrix $\mathbf{A}$ contains the mixing coefficients. The goal of BSS is to find an $M \times N$ demixing matrix $\mathbf{W}$ such that $M \times 1$ output vector

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$$\mathbf{u}(t) = \mathbf{W}\mathbf{x}(t) = \mathbf{W}\mathbf{A}\mathbf{s}(t) = \mathbf{P}\mathbf{D}\mathbf{s}(t) \quad (2)$$

where $\mathbf{P} \in \mathbf{R}^{M \times M}$ is a permutation matrix and $\mathbf{D} \in \mathbf{R}^{M \times M}$ is a diagonal scaling matrix.

When $N < M$, i.e., the number of the mixed signals is less than that of the source signals, BSS becomes a difficult case of the underdetermined problem. The complete separation is usually out of question in this case [14]. In our proposed method, we circumvent this problem by using the key signals to achieve the decryption.

III. Encryption

A block diagram of the BSS based speech encryption system is shown in Fig. 1. For encryption, there are two main stages, the segment splitter and the BSS mixing.

A. Segment Splitter

Similar to the time element scrambling [1], the original speech signal is firstly divided into equal time periods called frames. Each frame is then sub-divided into K ($K \geq 2$) smaller equal time periods called segments, which are denoted by $s_1(t), \ldots, s_K(t)$, $t = 1, \ldots, T$, where T is the segment length. Instead of permuting the segments within each of the frames as the time element scrambling does, the proposed system mixes the K speech segments within each frame with K statistically independent key signals (pseudo-random noise) in a specific way.

The proposed system encrypts the original speech frame by frame. For simplicity, we only discuss how to encrypt the present frame.

B. BSS Mixing

In this stage, the proposed system first generates K statistically independent key signals $s_{n1}(t), \ldots, s_{nK}(t)$, $t = 1, \ldots, T$, with random values uniformly distributed between -1 and 1, and a $K \times 2K$ mixing matrix $\mathbf{A}$, then mixes the K original speech segments $s_1(t), \ldots, s_K(t)$ and the K key signals $s_{n1}(t), \ldots, s_{nK}(t)$ by using the BSS mixing model (1), i.e., $\mathbf{s}(t) = [s_1(t), \ldots, s_K(t), s_{n1}(t), \ldots, s_{nK}(t)]^T$, $\mathbf{x}(t) = \mathbf{A}\mathbf{s}(t) = [x_1(t), \ldots, x_K(t)]^T$. K mixed speech segments $x_1(t), \ldots, x_K(t)$, $t = 1, \ldots, T$, are K encrypted speech segments. Finally, K encrypted speech segments are rejoined together to form one encrypted frame $x(t) = [x_1(t), \ldots, x_K(t)]$ for transmission.

To achieve the secure goal, we construct the mixing matrix $\mathbf{A}$ for encryption as follows:

$$\mathbf{A} = [\mathbf{B} \quad \beta\mathbf{B}] \quad (3)$$

where β is a scalar value, usually $1 \leq \beta \leq 3$. $\mathbf{B}$ is a $K \times K$ matrix of full rank, which are usually pseudo-randomly generated with values uniformly distributed between -1 and 1.

By using $\mathbf{A}$ in (3), BSS becomes the difficult case of the underdetermined problem where there are fewer mixed signals (K) than source signals ($2K$). As a result, the K original speech segments cannot be separated out from the K encrypted speech segments through BSS without the K key signals.

C. Segment Information

To correctly decrypt the encrypted speech received and recover the original speech, we need some prior knowledge about the segment splitter and the waveform of K original speech segments within each frame called the segment information. We mainly record the following information: the information about the segment splitter including the number of segments K within each frame and the segment length T; the waveform information including the zero cross count zcc, the maximum s_{max} and the minimum s_{min} of each original speech segment. For transmission, the segment information may be inserted into the encrypted speech file in a definite format, or transmitted together with the key signals by a secure channel.

IV. Decryption

The decryption part of the system consists of two main stages, the BSS decryption and the segment reconstruction, as shown in Fig. 1. The original speech will be recovered frame by frame.

A. BSS Decryption

In this stage, the first thing needed to do is to retrieve the K encrypted speech segments $x_1(t), \ldots, x_K(t)$ of each frame from the encrypted speech received by using the segment information K and T. Next, the BSS decryption separates the K original speech segments out from the K encrypted speech segments by means of the K key signals. This is similar to the decryption method used in [13]. The

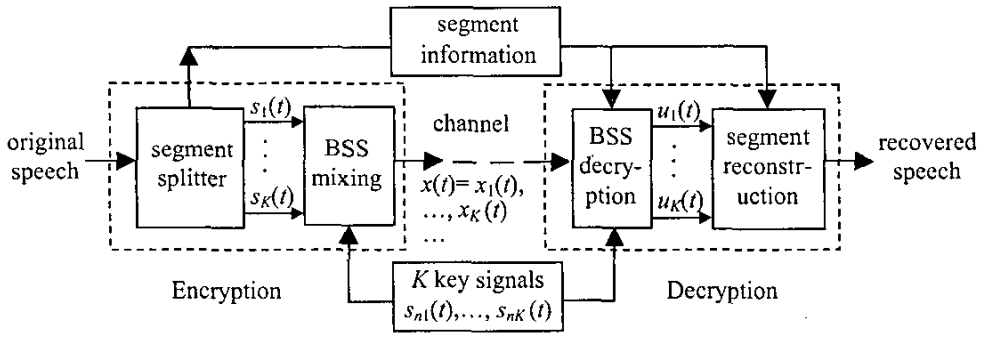


Fig. 1. Block diagram of proposed system.

K encrypted speech segments are firstly combined with the K key signals to form $2K$ mixed signals $\mathbf{x}(t) = [x_1(t),..., x_K(t), s_{n1}(t),..., s_{nK}(t)]^T$, on which the BSS algorithm are then performed. We used the infomax algorithm [9] in all the simulations reported below. Other popular BSS algorithms such as Comon's algorithm [8] and the fastICA algorithm [10] can also be used. When these well-known algorithms are compared, they generally perform near equally well.

It is easy to show that the equivalent mixing matrix for the BSS decryption is

$$\mathbf{A} = \begin{bmatrix} \mathbf{B} & \beta\mathbf{B} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}. \quad (4)$$

where $\mathbf{I}$ is a $K \times K$ identity matrix, $\mathbf{0}$ is a $K \times K$ zero matrix. Obviously, this $\mathbf{A}$ is a $2K \times 2K$ square matrix of full rank, and the underdetermined problem resulting from the BSS mixing encryption becomes the simplest BSS case, where the number of the mixed signals is equal to that of the source signals. Therefore, the complete separation of the source signals is available. The K original speech segments and the K key signals can be well recovered through the BSS decryption together, and the K decrypted speech segments, denoted by $u_1(t),..., u_K(t)$ (referring to Fig. 1), can be easily obtained by discarding the K key signals using *zcc*.

B. Segment Reconstruction

From (2), we can see that the K decrypted speech segments $u_1(t),..., u_K(t)$ are separated out with permutation and changes of scales and signs, compared with the K original speech segments $s_1(t),..., s_K(t)$ [7]. Therefore, we need to reconstruct the original speech from the decrypted speech segments by using the segment information.

First, the order of the K decrypted segments can be correctly rearranged by using the segment information *zcc*. Then, the waveform of them including the amplitudes and the signs can be readily reconstructed by using the segment information s_{max} and s_{min} of each original speech segment corresponding.

V. Computer Simulation Results

To verify the validity of the proposed system, extensive computer simulations are carried out. In most experiments, recorded speech files in wave-format are used and transmitted in file transfer protocol (FTP) way within local area network (LAN). Some other ways such as e-mail can also be used. One experiment is to securely transmit a speech file recording a person saying the digits "one" to "ten" in English. It is one of the separated speech signals in the experiments of [15]. A little music interference is audible. The speech signal is sampled at 16KHz and 6.875 seconds long (110000 samples), as shown in Fig. 2.

We experiment with many different K and T to test the performance of the proposed system. In the following, we shall take two cases of them as examples.

A. Case 1

Let K=4 and T=27500, i.e., the original speech signal is treated as one frame, which is sub-divided into 4 segments $s_1(t)$, ... , $s_4(t)$ with 27500 (about 1.72 seconds) samples. Fig. 3(a) shows the waveform of the 4 original speech segments. Besides the information K and T, the proposed system records the segment information about the waveform of the 4 original speech segments, as shown in Table I.

Next, the proposed system generates 4 statistically independent key signals $s_{n1}(t)$, ... , $s_{n4}(t)$, t = 1, ... , 27500, with random values uniformly distributed between -1 and 1 shown in Fig. 3(b), and a 4×8 mixing matrix $\mathbf{A}$ according to (3), where β=2 and one example of $\mathbf{B}_{4\times4}$ is

$$\mathbf{B}_{4\times4} = \begin{bmatrix} -0.0928 & 0.2057 & 0.2148 & 0.9920 \\ 0.8080 & 0.5212 & -0.7685 & 0.4909 \\ 0.9031 & -0.7282 & 0.5214 & 0.9798 \\ 0.2041 & -0.6465 & 0.4807 & -0.2712 \end{bmatrix}.$$

Then, the proposed system mixes $s_1(t)$, ... , $s_4(t)$ and $s_{n1}(t)$, ... , $s_{n4}(t)$ by using the BSS mixing model (1). Fig. 3(c) shows four encrypted speech segments $x_1(t)$, ... , $x_4(t)$, t = 1, ... , 27500. They all sound like white noise, and have reached the security goal. Finally, the encrypted speech signal for transmission is formed by rejoining the four encrypted speech segments together.

On the receiving side, the four encrypted speech segments $x_1(t)$, ... , $x_4(t)$ of Fig. 3(c) are retrieved by using the segment information K and T, and then combined with the four key signals $s_{n1}(t)$, ... , $s_{n4}(t)$ of Fig. 3(b) to form eight mixed signals for the BSS decryption. Fig. 4(a) shows one example of four decrypted speech segments $u_1(t)$, ... , $u_4(t)$ before segments reconstruction. They are all of good quality, but in different orders and with changed amplitudes and signs, compared with the four original speech segments of Fig. 3(a). Fig. 4(b) shows four recovered speech segments after the segment reconstruction according to the segment information shown in Table I. The original speech signal is thus recovered by rejoining the four recovered speech segments together.

B. Case 2

Let K=8 and T=13750, i.e., the original speech signal is still treated as one frame, but each frame is sub-divided into 8 segments $s_1(t)$, ... , $s_8(t)$ with 13750 (about 0.86 seconds) samples. Their waveform and waveform information are shown in Fig. 5(a) and Table II, respectively. Fig. 5(b) shows 8 statistically independent key signals $s_{n1}(t)$, ... , $s_{n8}(t)$, t = 1, ... , 13750. The 8×16 mixing matrix $\mathbf{A}$ is similarly generated as that in Case 1 according to (3) with β=1. Fig. 5(c) shows eight encrypted segments $x_1(t)$, ... , $x_8(t)$, t = 1, ... , 13750, having reached the security goal. The eight encrypted speech segments are rejoined together to form the encrypted speech signal for transmission in the end.

On the receiving side, the eight encrypted speech

segments $x_1(t)$, ... , $x_8(t)$ of Fig. 5(c) are retrieved by using the segment information K and T, and then combined with the eight key signals $s_{n1}(t)$, ... , $s_{n8}(t)$ of Fig. 5(b) to form sixteen mixed signals for the BSS decryption. Fig. 6(a) shows one case of eight decrypted speech segments $u_1(t)$, ... , $u_8(t)$ with changed order, amplitudes and signs. Fig. 6(b) shows eight recovered speech segments after the segment reconstruction according to the segment information shown in Table II. The original speech signal is then recovered by rejoining the eight recovered speech segments together.

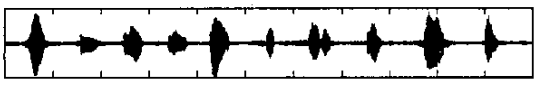


Fig. 2. Original speech signal (spoken digits 1-10 with a little music interference) with 110000 samples.

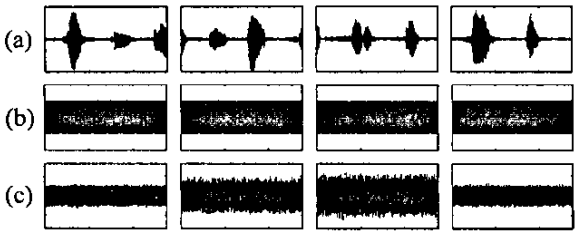


Fig. 3. Example of encryption in Case 1 (K=4, T=27500). (a) Four original speech segments $s_1(t)$, ... , $s_4(t)$. (b) Four key signals $s_{n1}(t)$, ... , $s_{n4}(t)$. (c) Four encrypted speech segments $x_1(t)$, ... , $x_4(t)$.

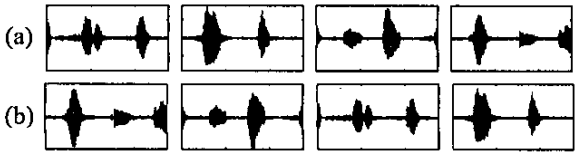


Fig. 4. One example of four decrypted speech segments in Case 1 with infomax algorithm. (a) Four decrypted speech segments $u_1(t)$, ... , $u_4(t)$ before segments reconstruction. (b) Four recovered speech segments after segments reconstruction.

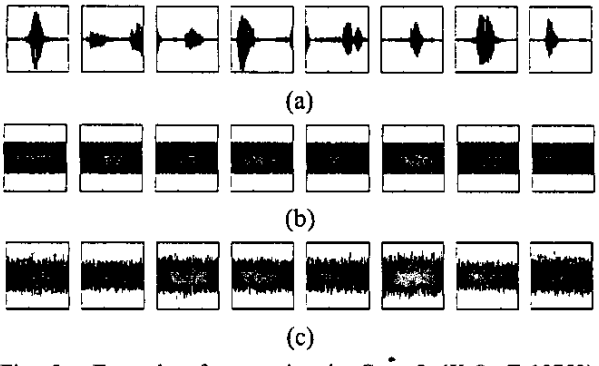


Fig. 5. Example of encryption in Case 2 (K=8, T=13750). (a) Eight original speech segments $s_1(t)$, ... , $s_8(t)$. (b) Eight key signals $s_{n1}(t)$, ... , $s_{n8}(t)$. (c) Eight encrypted speech segments $x_1(t)$, ... , $x_8(t)$.

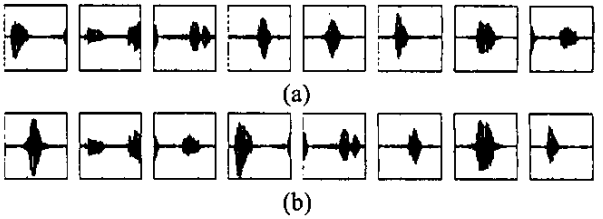


Fig. 6. One example of eight decrypted speech segments in Case 2 with infomax algorithm. (a) Eight decrypted speech segments $u_1(t)$, ..., $u_8(t)$ before segments reconstruction. (b) Eight recovered speech segments after segments reconstruction.

TABLE I

Waveform Information about 4 Original Speech Segments in Case 1

	$s_1(t)$	$s_2(t)$	$s_3(t)$	$s_4(t)$
zcc	4315	5905	7456	5239
s_{max}	0.8594	0.7306	0.5633	0.8795
s_{min}	-0.9597	-1.0002	-0.5514	-0.7750

TABLE II

Waveform Information about 8 Original Speech Segments in Case 2

	zcc	s_{max}	s_{min}
$s_1(t)$	2290	0.8600	-0.9592
$s_2(t)$	2019	0.4785	-0.5480
$s_3(t)$	2364	0.3493	-0.4630
$s_4(t)$	3533	0.7305	-1.0004
$s_5(t)$	4340	0.5635	-0.4710
$s_6(t)$	3100	0.5550	-0.5515
$s_7(t)$	2260	0.8796	-0.7749
$s_8(t)$	2989	0.7757	-0.5736

C. Performance Analysis

To quantify the performance of the proposed algorithm, we computed the signal-to-noise ratio (*SNR*) index for $s_i(t)$, the ith component of $\mathbf{s}(t)$, in the recovered signals by the BSS decryption, which is defined as follows:

$$SNR_i = 10\log_{10}\frac{\sum_{t=1}^{T}\left[(\mathbf{WA})_{ii}s_i(t)\right]^2}{\sum_{j=1}^{M}\sum_{t=1}^{T}\left[(\mathbf{WA})_{ij}s_j(t)\right]^2}, \quad i \neq j \qquad (5)$$

where $\mathbf{W}$ is the demixing matrix gotten from the BSS decryption algorithm used. $\mathbf{A}$ is the mixing matrix corresponding to the mixed signals for the BSS decryption similar to (4). $(\mathbf{WA})_{ij}$ denotes the ith row and the jth column of matrix $\mathbf{WA}$. In (5), the order of the recovered signals is assumed to be the same as that of the source signals, which can be done by permuting the rows of $\mathbf{WA}$.

SNR_i was computed for the K original speech segments (Fig. 3(a) and Fig. 5(a)) in the K decrypted speech segments (Fig. 4(b) and Fig. 6(b)) for Case 1 and Case 2, respectively. Table III and Table IV show the satisfactory results. We can find that the SNR_i indexes are all very high. The reason is that the proposed system separates the key signals out by the BSS decryption with very high accuracy. Thus, the voice quality of the recovered speech is excellent.

TABLE III

SNR_i (dB) of Four Original Speech Segments in Four Decrypted Speech Segments in Case 1

$s_1(t)$	$s_2(t)$	$s_3(t)$	$s_4(t)$
75.9	78.2	92.5	86.2

TABLE IV

SNR_i (dB) of Eight Original Speech Segments in Eight Decrypted Speech Segments in Case 2

$s_1(t)$	$s_2(t)$	$s_3(t)$	$s_4(t)$	$s_5(t)$	$s_6(t)$	$s_7(t)$	$s_8(t)$
76.4	63.2	58.7	74.8	72.8	70.7	68.0	76.7

VI. CONCLUSTION

We have described a BSS based speech encryption system. The results of computer simulation illustrate the proposed system has a high level of security and excellent audio quality.

There are two main aspects for the proposed system to ensure its security. First, the proposed system encrypts multiple original speech segments within each frame simultaneously instead of treating every segment of speech independently. This makes multiple original speech segments cover mutually through mixing among themselves while the key signals cover them. As a result, the encrypted speech sounds like white noise. With the BSS accurate decryption, the proposed system, however, can still achieve excellent audio quality for the recovered speech by separating the key signals out from the encrypted speech segments. Second, the proposed system can generate statistically independent key signals for different original speech segments, and does not impose any restriction on the key space. Therefore, the proposed system has a very high level of security, and is suitable for transmitting and storing the secure speech files.

ACKNOWLEDGEMENT

This work was supported by the National Natural Science Foundation of China under Grant No. 60172073 and No. 60372082, Trans-Century Training Program Foundation for the Talents by the Ministry of Education of China, and Foundation for University Key Teachers by the Ministry of Education of China.

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